Kenneth Lance Howard

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Education

North Carolina A & T State University, B.S. in Computer Science- **Magna Cum Laude** **Greensboro, NC | May 2025**
North Carolina A & T State University, M.S. in Computer Science-**AI and DS focus GPA: 4.0 Greensboro, NC | Exp. Dec 2026**

Technical Skills & Software Proficiencies

Proficient: Python, Java, C++, Swift, JavaScript, React, SQL, Linux, Node.js, APIs/microservices, PyTorch, Kubernetes
Skills & Interests: Fullstack Development, Mobile & Web Applications, Distributed Systems, Data Structures & Algorithms, Machine Learning, AI/ML Research, RAG, Cloud Services, Deep Learning.

Work Experience

Sandia National Laboratories **May 2024 – Present**
Computer Science Research & Development - Nuclear Deterrence **Albuquerque, NM**
Conducting national security research on advanced algorithms and secure AI systems.
- Improved data pipeline efficiency by nearly **50%** through algorithm design.
- Integrated **LLMs**, **RAG**, and image models to streamline workflows for several national security projects.
- Partnered with cross-functional teams to deliver mission-critical solutions.

Graduate Research in AI/ML **Aug 2024 – Present**
AI Full-Stack Developer – NCAT Research Project **Greensboro, NC**
Building AI applications for education with intelligent assistants and scalable deployment.
- Built an AI assistant actively used by students and faculty.
- Developed full-stack features with **Streamlit**, **LangChain**, and **Docker**.
- Led backend integration to support scalability and adoption.

Boeing Research Collaboration **Aug 2025 – Present**
Graduate Researcher – 3D Modeling & Predictive AI Systems **Greensboro, NC**
Exploring AI for aerospace with Vision-Language Models and predictive simulations in Unity.
- Prototyped **VLMs in C#**, improving decision-making accuracy by at least **20%**.
- Built Unity simulations processing **1,000+ real-time data points** per run.
- Applied generative AI to strengthen predictive modeling and testing.

Projects

- **Full Stack Application Development:**
  *Completed large-scale **Java** projects with **front-end/back-end** design using **Eclipse**, **OOP**, and **Jenkins**; built apps in **React**, **React Native**, and **Python** with API integrations, scripts, and web scraping.*
  - Designed architecture, implemented features, integrated **RESTful APIs**, and ensured real-time data sync.
- **AI Data Visualization Platform – J.P. Morgan Data for Good** (**React**, **Tailwind**, **OpenAI API**, **Python**):
  *Built a **full-stack** web app that visualized NBA player stats and mental-health trends from CSV data using **LLMs**.*
  - Engineered the front end with **React**/**Tailwind**; generated charts/summaries from natural-language prompts via **OpenAI API**.
- **Capital One Custom AI Assistant (1st Place)**:
  *Built an assistant leveraging **LLMs** with **Streamlit** and **Docker** for contextual responses, data retention, and structured insights.*
  - Used model development, testing, and containerized deployment for reliable performance.
- **Deutsche Bank Hackathon — Sentiment Analysis**:
  *Developed a real-time pipeline to classify financial news and social media data with **NLP models** integrated into a live dashboard.*
  - Trained advanced **NLP models** and integrated results into a live analytics dashboard.
- **Sonify – Hearing Health Assistant**:
  *Developed a mobile app in **Flutter**, integrating AI-driven music suggestions and hearing health insights.*

Leadership & Involvement

- **Python Bootcamp Leader**, NC A&T: Led sessions to strengthen coding fundamentals for peers **Spring 2024**
- **Woodland Ellroy Hall Fellowship Recipient (1 of 15 selected)**: Recognized for academic excellence and leadership **2025**
- **Microsoft Mentorship Program (1 of 10 selected)**: Focused on Copilot, CS, and personal development **Fall 2024**
- **Dell Generative AI Certification & Award**: Earned industry-recognized credential in applied AI **Fall 2024**
- **Hackathon Leadership:** Multiple 1st Place finishes (Chevron, Capital One, Deutsche Bank) **Fall 2023–Fall 2024**
- Member, **ACM, NSBE, and SHPE**, NC A&T **Fall 2023–Present**
- Member, **NCSU AI Club** **Fall 2021–2022**